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DISTRIBUTION UTILITY MEET
DUM 2026

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DUM 2026 CONFERENCE SESSION

SESSION 4: FLEXIBILITY SOLUTIONS FOR DISTRIBUTION NETWORKS AND CEA STANDARDS FOR SOLAR PV AND BESS

Making Flexibility the Core Resource of Distribution Grids

Presented By

Disha Khosla, AGM – Technical Services, ISGF

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



A RENEWABLE-RICH, DECENTRALISED GRID

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Over half of India's capacity is now non-fossil, and more of it sits on distribution networks

>50%
of installed generation capacity is non-fossil

165 GW
solar capacity, with rooftop solar growing under PM Surya Ghar

58 GW
wind capacity

75 million
smart meters already installed under RDSS

174 GW
energy storage needed by 2035 to back the transition

Yesterday: one-way distribution

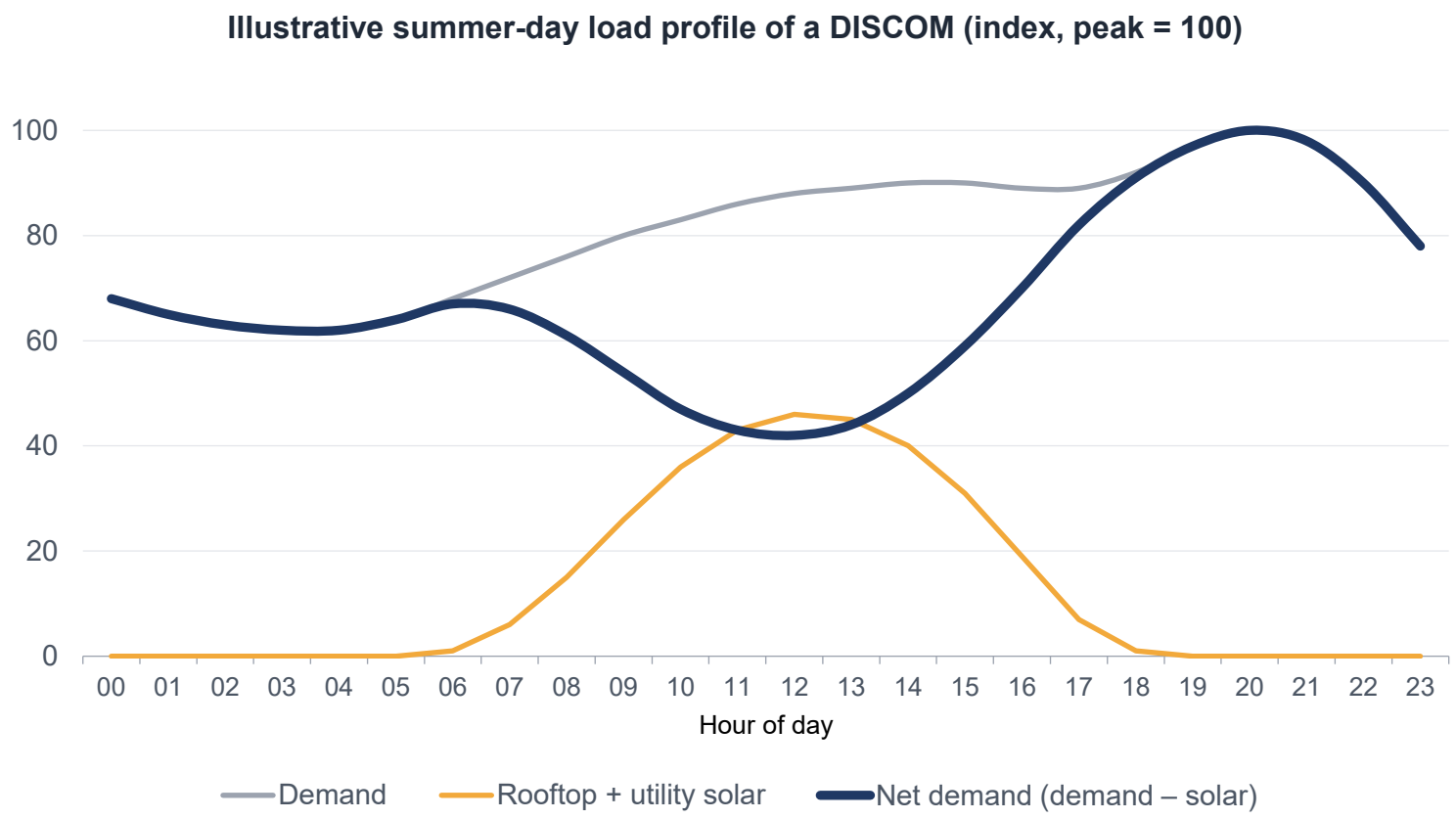
- Power flows one way, from substation to consumer
- Predictable, slowly growing load
- Peaks met by adding network and generation capacity

Today: multi-directional networks

- Rooftop solar and batteries inject and draw power; soon EVs will also be able to inject power to the grid
- Daytime surplus and reverse power flows on LV/MV feeders
- Steep evening ramps and short, sharp summer peaks

THE DUCK CURVE REACHES DISTRIBUTION

A midday solar surplus, then a steep evening ramp as solar output falls to zero



1 Midday trough

Net demand drops sharply. Excess solar pushes up voltages and reverses power flow on LV/MV feeders.

2 Evening ramp

Demand rises just as solar falls away. DISCOMs must buy costly peak power within 2–3 hours.

3 Summer peaks

Cooling-driven peaks are growing fastest in all major load centres.

DELHI AND MUMBAI PEAK DATA

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Annual hours at or above share of peak, 2021 – Sep 2026 (Delhi 5-min, Mumbai hourly)

City	Period	Peak at	Peak MW	≥99%	≥95%	≥90%	≥75%	Days ≥95%	Days ≥90%	Load factor %	Coverage %
Delhi	CY2021	2/7 15:15	7,292	1.2	6.7	29	381	4	8	48.1	93.1
Delhi	CY2022	29/6 15:05	7,673	0.9	8.5	53	711	6	19	52.1	92.5
Delhi	CY2023	22/8 15:15	7,410	0.8	15.2	74.2	1,050	9	31	53.3	97.5
Delhi	CY2024	19/6 15:05	8,632	1.3	7.7	48.3	666	4	19	50.8	95.8
Delhi	CY2025	12/6 23:10	8,419	2.3	9.8	29.9	564	3	7	51.3	98.6
Delhi	Jan–Sep 2026	29/6 15:15	8,705	0.8	13.2	72.2	817	6	25	56.1	94.7
Mumbai	CY2021	6/10, hour 16	3,270	4	71	262	2,853	19	56	69.2	99.5
Mumbai	CY2022	28/4, hour 16	3,851	3	19	151	2,234	6	39	65	97.8
Mumbai	CY2023	9/6, hour 16	4,108	1	10	139	2,140	7	37	65.6	97.5
Mumbai	CY2024	21/5, hour 16	4,306	2	21	119	1,936	11	30	64.7	99.2
Mumbai	CY2025	9/4, hour 16	4,164	1	17	213	2,596	12	37	67.5	98.9
Mumbai	Jan–Sep 2026	8/6, hour 16	4,631	4	53	232	1,874	22	48	-	-

Five-year average 2021–2025		
	Delhi	Mumbai
≥99% of Peak	1.3 Hr	2.2 Hr
≥95% of Peak	9.6 Hr	27.6 Hr
≥90% of Peak	46.9 Hr	177 Hr

CITY PEAK ESTIMATES

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In most cities the top 5% of peak load occurs for under 100 hours a year (indicative)

City (data basis)*	Peak reference	≥99% ("at max")	≥95% (top 5%)	≥90% (top 10%)	≥75% (top 25%)	Confidence of Estimates by AI Model
Tamil Nadu (calculated)	15,361 MW (2017) 21,724 MW (2026)	~20 h	~190 h	~730 h	~5,500 h	Medium-high
Chennai city (inferred from TN data)	—	~5–10 h	~60–120 h	~300–500 h	~3,000–4,000 h	Low
Delhi (anchored on SLDC data)	8,656 MW (2024)	~3–5 h (86 MW)	~15–30 h (432 MW)	~70–120 h (865 MW)	~700–1,100 h	Medium
Kolkata (CESC)	~2,200 MW in old planning figures	~3–5 h	~20–40 h	~100–200 h	~1,000–1,500 h	Low
Mumbai (SLDC)	~2,300 MW (AEML) 4642 MW (all of Mumbai in 2026)	~5–10 h	~40–80 h	~200–400 h	~2,500–3,500 h	Low
Bengaluru (BESCOM)	~5,900 MW (BESCOM, 2019)	~3–8 h	~30–60 h	~150–300 h	~2,000–3,000 h	Low
Hyderabad / Pune	—	~3–8 h	~30–60 h	~150–300 h	~1,500–3,000 h	Low

WHY NETWORK UPGRADES ALONE FALL SHORT

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With DISCOM finances a top priority, flexibility must become a core resource

Build for the peak (traditional)

- Upgrade transformers, feeders and substations for peaks that last under 100 hours a year
- Buy expensive short-term power at peak
- Assets idle most of the year; costs flow into tariffs
- Slow: years to plan, approve and build

Manage the peak (flexibility-first)

- Shift, store and shave demand with DR, ToD tariffs, BESS and EV smart charging
- Non-wire alternatives defer network capex
- Data from 75 million smart meters targets the right feeders and consumers
- Fast to deploy and scales with need

Cooling is the main driver of demand growth across India**. Strategies must tackle AC load directly: district cooling, thermal storage and smart AC controls**.

FLEXIBILITY ACROSS THE POWER SYSTEM

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A broad toolbox exists at every level: generation, transmission and distribution

GENERATION

- **Flexible coal plants:** lower minimum load (55% to 40%), faster ramps
- **Pumped storage hydro:** multi-hour shifting, inertia
- **RE + co-located storage:** firm, dispatchable solar and wind (FDRE, RTC)
- **Hydro and gas peaking:** fast support for evening peaks
- **Markets:** RTM and ancillary services reward ramping

TRANSMISSION

- **Grid-scale BESS:** peak shifting, frequency, congestion relief
- **Grid-forming inverters:** voltage and frequency support, black start
- **Dynamic line rating:** more capacity from existing lines
- **STATCOM / FACTS:** voltage control and system strength
- **Inter-regional transfers:** share RE surpluses across states

DISTRIBUTION

- **Demand response:** and dynamic ToD tariffs
- **Neighbourhood BESS:** at DTs, plus BESS for new large loads
- **EV smart charging and V2G:** midday charging, evening discharge
- **District cooling + thermal storage:** MW-scale load shifting
- **Smart inverters, VPPs, aggregators:** visible, dispatchable DERs

This session focuses on distribution: closest to consumers, and where DISCOMs must become active system managers

TAMING THE DUCK CURVE

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Shift, Store, See and Steer before spending on new wires

SHIFT

Move demand into solar hours

- Solar-hour ToD discounts
- Daytime farm supply via PM-KUSUM feeder solarisation
- Schedule water heating, pumping and EV charging at midday

STORE

Bank the midday surplus

- DT-level and neighbourhood BESS: charge at noon, discharge at the evening peak
- BESS for new large loads
- Chilled-water and ice thermal storage

SEE

Know where the stress is

- Smart-meter and DER visibility (National SRTPV Registry)
- Feeder- and DT-level forecasting
- Hosting-capacity maps for rooftop solar

STEER

Control at the grid edge

- Smart-inverter volt-var and volt-watt settings
- DERMS/ADMS orchestration and VPP dispatch
- Export limits; curtailment only as a last resort

Outcome: a flatter net load, less peak power to buy, and network capex deferred

GRID-FORMING INVERTERS FOR BESS

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CEA now requires grid-forming inverters for new BESS of 50 MW and above

	Grid-following (GFL)	Grid-forming (GFM)
Behaviour	Follows grid voltage (current source)	Sets its own voltage and frequency (voltage source)
Grid strength	Needs a strong grid	Stable in weak grids (SCR ≤ 2.0)
Inertia / fault current	Little or none	Synthetic inertia; 1.5 pu for 200 ms
Black start	No	Yes, mandatory at 50 MW and above

CEA Advisory - 10 Jul 2026

- 100% GFM inverters for BESS of 50 MW and above
- Response starts within 5 ms; full response within 30 ms
- Rides through ±60° phase jumps and high RoCoF
- Damps oscillations from 0.1 to 20 Hz; 24×7 AGC participation
- Draft CEA Connectivity Standards 2026: LVRT/HVRT and plant controllers for solar PV and BESS

A new technology: what DISCOMs, TRANSCOs and customers must learn

- EMT studies and model validation at the design stage
- Protection settings for new fault-current behaviour
- Commissioning tests and periodic re-verification by LDCs
- How many GFM and GFL units interact on weak networks
- Training for planners, protection engineers and operators
- Procurement specs and type testing for smaller BESS

SMART-METER DATA FOR DSM

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75 million RDSS smart meters make every consumer a source of grid insight

75 million

smart meters installed under RDSS

15–30 min

interval data for every consumer

AI and analytics turn this data into targeted, verifiable flexibility

Load forecasting	Feeder, DT and segment forecasts of peak size, timing and duration
Consumer segmentation	Find flexible loads (ACs, pumps, EVs) and target DR offers, with consent and privacy safeguards
DT overload alerts	Early warning of overloaded DTs and feeders to prevent local blackouts
Rooftop solar visibility	Detect net export and reverse flow; feed hosting-capacity analysis
DR baselines and M&V	Measure delivered flexibility for settlement and incentives
Tariff design	Test price response to ToD and design dynamic tariffs



DYNAMIC TIME-OF-USE TARIFFS

From static ToD slabs to prices that follow the grid

1 Flat tariff

Same price at all hours

INDIA TODAY

2 Static ToD

Fixed solar-hour discounts and peak premiums

3 Seasonal ToD + peak rebates

Critical-peak pricing and peak-time rebates on event days

4 Day-ahead dynamic

Prices linked to power-exchange prices, sent a day ahead

5 Real-time, automated

Price signals drive smart ACs, EV chargers and BESS

ToD rules in force (Rights of Consumers Amendment, 2023)

- Solar hours (8 h/day): at least 20% below the normal tariff
- Peak hours: at least 1.2× normal for C&I, 1.1× for others
- C&I ≥10 kW from Apr 2024; others (except agriculture) from Apr 2025; immediately on smart-meter installation

Design principles for dynamic tariffs

- Opt-in, with bill protection in the first year
- Day-ahead price signals via smart meter, app and SMS
- Automation so consumers need not act by hand
- Clear SERC rules on price bands and settlement

DISTRICT COOLING FOR LARGE DEVELOPMENTS

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Cooling drives demand growth, so tackle it at the source

Individual AC units

- Every flat and office runs its own split or window ACs
- Cooling demand peaks with the grid peak
- No storage: each kW of cooling is a kW at peak
- Estimated Cooling Demand by 2040: 180 GW

District Cooling System (DCS)

- A central chiller plant serves a campus or colony through a chilled-water network
- Insulated chilled-water tanks store cooling made in off-peak or solar hours
- Stored cooling serves the evening peak
- Cooling can be provided as a Service against Monthly Fee - like electricity, water and gas

>50%

lower power demand than individual AC units

Plus MW-scale flexibility for the DISCOM from thermal storage

The ask: mandate DCS with thermal storage

- Large campuses and institutions
- Commercial complexes and IT parks
- Residential colonies and townships

- Write DCS into building codes and master plans
- Size thermal storage for the evening peak
- ToD tariffs that reward shifting cooling load
- **Need a Regulator for the Cooling Sector – till then allow SERCs to regulate DCS**

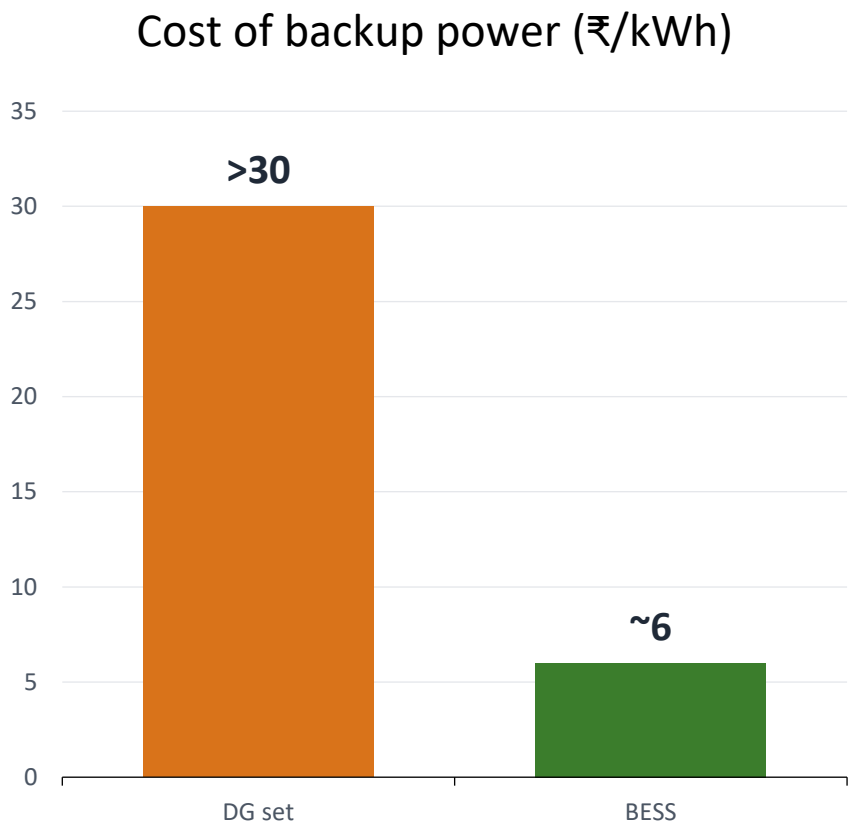
- India's first DCS in GIFT City Gujarat in 2014.
- ISGF White Paper on DCS in 2020 and BEE's Technical Guidelines for DCS in 2023 pushed the DCS agenda in India – today there are 30+ functional DCS; and over 100 projects in progress

2-HOUR BESS FOR NEW LOADS >100 kW

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Replace idle diesel standby with batteries that serve the grid 24×7



Why DG sets are a dead asset

- Used only during outages, now nil or a few hours a year in most cities
- Power costs over ₹30/kWh, plus emissions and noise

ISGF proposal for Regulators and DISCOMs

- **Mandate at least 2 hours of BESS for new connections above 100 kW (e.g. 200 kWh for a 100 kW load)**
- **Grid-connected BESS gives backup plus flexibility services 24×7**
- **Customers cut bills through ToD arbitrage and lower peak-demand charges**
- **SERCs to offer incentives for flexibility services from customer BESS**

MAKING FLEXIBILITY A TRADABLE ASSET

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Regulation must let flexibility be measured, paid for and traded

1 RECOGNISE

- State DF/DSM regulations that recognise aggregators (e.g. Rajasthan, 2026)
- Flexibility planned alongside network capex

2 MEASURE

- Standard baselines and M&V from smart-meter data
- Telemetry and DER registries (e.g. National SRTPV Registry)

3 PAY

- Capacity plus utilisation payments; peak-time rebates
- Share network-deferral value; allow costs in ARR

4 TRADE

- Local flexibility markets run by DISCOMs
- Aggregated access to RTM and ancillary services; value stacking

Make it bankable: VGF and green finance for distribution-level BESS, plus standard contracts that give investors predictable revenue

SCALING VPPs AND AGGREGATORS

Thousands of small assets, dispatched as one power plant

- **Rooftop solar**
- **Home and C&I BESS**
- **EVs and V2G chargers**
- **Smart ACs and DCS**
- **Agricultural pumps**

Distributed assets

Aggregator / VPP platform

- Forecast flexibility
- Optimise across prices and grid limits
- Dispatch in real time
- Measure, verify and settle

- **DISCOM:** peak and congestion relief
- **SLDC:** ancillary services and balancing
- **Power exchange:** RTM / DAM trading
- **Consumers:** bill savings and payments

Enablers: open standards (OpenADR, IEEE 2030.5, OCPP) · telemetry to DISCOM and SLDC · minimum bid sizes aggregators can meet · cybersecurity and consent rules

KEY TAKEAWAYS AND RECOMMENDATIONS

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DISCOMs must move from passive distributors to active system managers

1. Treat flexibility as a core resource: non-wire alternatives before network upgrades
2. Prepare for grid-forming BESS: EMT studies, protection updates, testing and training
3. Use RDSS smart-meter data for forecasting, DR targeting and M&V
4. Move from static ToD to dynamic TOU and automation-friendly tariffs
5. Mandate district cooling systems with thermal storage for large developments
6. Mandate 2-hour BESS for new loads above 100 kW, with incentives for flexibility services
7. Build regulatory and financial frameworks that pay for and trade flexibility
8. Scale VPPs and aggregators with open standards and market access

THANK YOU

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